Class 6 Science – Chapter 2: Diversity in the Living World

Complete Question–Answer Notes | Simple English | CBSE

This chapter explains biodiversity, classification of plants and animals, habitats, and adaptations that help organisms survive in different environments. The notes below follow the topics and activities in your uploaded 12-page chapter PDF, including the pictures, tables and "Think and Answer" activities.

02-diversity-in-the-living-world.pdf

1. All questions and answers from the chapter

A. Exploring Habitat — Page 1

Q1. What is the name of the given habitat?

Answer: The given habitat is a desert.

Q2. Name the animals that may be seen in a documentary about a desert.

Answer: Camels, snakes, lizards, scorpions, desert foxes and desert rats may be seen in a desert.

Q3. Name the plants that may be seen in a desert.

Answer: Cactus, thorny shrubs and other plants adapted to dry conditions may be seen in a desert.

Q4. Can a dog survive in that place? Give reasons for your answer.

Answer: A dog may survive in a desert for a limited time if it gets enough water, food and shelter. However, extreme heat and a shortage of water can make survival difficult.

B. Think and Answer — Page 7

Q5. Can an organism live in two different habitats? If yes, give an example.

Answer: Yes, some organisms can live in two different habitats. For example, a frog can live both on land and in water.

C. Identify These Animals — Page 12

The pictures show a camel (Animal A) and a fish (Animal B).

Q6. Image A shows a _____ and image B shows a _____.

Answer: Image A shows a camel and image B shows a fish.

Q7. Write the names of the habitats these animals are commonly found in.

Answer:

- Camel: Desert habitat.
- Fish: Aquatic habitat, such as ponds, lakes, rivers or seas, depending on the species.

Q8. List the adaptations of animal A.

Answer: A camel has the following adaptations:

1. Long legs help keep its body away from the hot sand.
2. Broad, padded feet help it walk on sand.
3. Long eyelashes and closable nostrils help protect it from blowing sand.
4. It produces a small amount of urine and loses little water.
5. Its hump stores fat, which serves as a food reserve.

Q9. What is the special shape of animal B? How does this shape help it survive in its habitat?

Answer: A fish has a streamlined body. This shape helps it move easily through water.

Q10. List the adaptations of animal B.

Answer:

1. Its streamlined body helps it swim easily.
2. Its fins help it swim and change direction.
3. Its gills help it breathe underwater.
4. Its tail helps it move forward and change direction.

2. Important questions and answers from every topic

Topic 1: Diversity in Plants and Animals Around Us

Q11. What is biodiversity?

Answer: Biodiversity is the variety of plants, animals and other living organisms found in the environment.

Q12. Why is biodiversity important?

Answer: Biodiversity is important because every organism plays a role in nature. Plants provide food and shelter, while animals help in pollination, seed dispersal and maintaining the balance of nature.

Q13. What is meant by diversity in the living world?

Answer: Diversity in the living world means the presence of different kinds of plants, animals and other living organisms.

Q14. How do plants and animals depend on each other?

Answer: Plants provide food and shelter to animals. Animals help plants through pollination and seed dispersal. In this way, plants and animals depend on each other.

Topic 2: How to Group Plants and Animals

Q15. Why do we classify plants and animals?

Answer: We classify plants and animals to organise them into groups based on their similarities and differences. This makes them easier to study and understand.

Q16. What is classification?

Answer: Classification is the process of grouping organisms according to their common features and differences.

Q17. Name some features used to classify plants.

Answer: Plants can be classified according to their height, stem, leaf venation, root system and seed type.

Q18. Name some features used to classify animals.

Answer: Animals can be classified according to their body shape, size, structure, colour, habitat and movement.

Topic 3: Classification of Plants Based on Their Height

Q19. What are herbs? Give examples.

Answer: Herbs are short plants with soft, green stems. They usually have few branches.

Examples: Coriander, mint, wheat and mustard.

Q20. What are shrubs? Give examples.

Answer: Shrubs are medium-sized, bushy plants with woody stems and branches arising near the base.

Examples: Rose, hibiscus, lemon and lantana.

Q21. What are trees? Give examples.

Answer: Trees are tall plants with a hard, thick, woody main stem called a trunk.

Examples: Neem, peepal, mango and jamun.

Q22. Differentiate between herbs, shrubs and trees.

Feature	Herbs	Shrubs	Trees
Height	Usually short	Medium-sized	Usually tall
Stem	Soft and green	Woody and hard	Thick, hard and woody
Branches	Few or small	Arise near the base	Usually arise higher up
Examples	Coriander, mint	Rose, hibiscus	Neem, mango

Topic 4: Classification of Plants Based on Their Stem

Q23. What are creepers? Give examples.

Answer: Creepers are plants with weak stems that spread along the ground because they cannot stand upright.

Examples: Watermelon and pumpkin.

Q24. What are climbers? Give examples.

Answer: Climbers are plants with weak stems that grow upwards with the support of other objects.

Examples: Money plant and pea plant.

Q25. Differentiate between creepers and climbers.

Answer: Creepers spread along the ground, while climbers grow upwards with the support of other objects.

Q26. What are tendrils?

Answer: Tendrils are thin, thread-like structures that help climbing plants attach themselves to a support.

Topic 5: Classification of Plants Based on Leaf Venation

Q27. What is venation?

Answer: Venation is the arrangement of veins and smaller veinlets in a leaf.

Q28. What is reticulate venation?

Answer: Reticulate venation is a pattern in which the veins form a network on the leaf.

Examples: Mango and peepal.

Q29. What is parallel venation?

Answer: Parallel venation is a pattern in which the veins run parallel to one another.

Examples: Grass and banana.

Q30. How can leaf venation help us identify a plant's root system?

Answer: In the general pattern described in this chapter, plants with reticulate venation usually have taproots, while plants with parallel venation usually have fibrous roots.

Topic 6: Classification of Plants Based on Their Roots

Q31. What is a root?

Answer: A root is the part of a plant that usually grows underground and helps hold the plant firmly in the soil.

Q32. What are the two main types of root systems?

Answer: The two main types are the taproot system and the fibrous root system.

Q33. What is a taproot? Give examples.

Answer: A taproot system has one main root that grows downwards, with smaller side roots branching from it.

Examples: Mustard, carrot and radish.

Q34. What is a fibrous root system? Give examples.

Answer: A fibrous root system has many thin roots of similar size arising from the base of the stem.

Examples: Wheat, rice and grass.

Q35. Differentiate between taproots and fibrous roots.

Taproot system	Fibrous root system
One main root is present.	Many thin roots are present.
Smaller roots branch from the main root.	Roots are usually similar in size.
Example: Carrot	Example: Wheat

Topic 7: Classification of Plants Based on Their Seeds

Q36. What is a cotyledon?

Answer: A cotyledon is a seed leaf present inside a seed.

Q37. What are monocot plants? Give examples.

Answer: Monocot plants have seeds with one cotyledon. They generally have parallel venation and fibrous roots.

Examples: Rice, wheat and maize.

Q38. What are dicot plants? Give examples.

Answer: Dicot plants have seeds with two cotyledons. They generally have reticulate venation and taproots.

Examples: Pea, chickpea and mango.

Q39. Differentiate between monocots and dicots.

Feature	Monocots	Dicots
Cotyledons	One	Two
Leaf venation	Usually parallel	Usually reticulate
Root system	Usually fibrous	Usually taproot
Examples	Rice, wheat	Pea, chickpea

Topic 8: How to Group Animals

Q40. How do animals move from one place to another?

Answer: Animals move by walking, jumping, flying, swimming, crawling and climbing.

Q41. Name the body parts used by different animals for movement.

Movement	Examples	Body parts used
Walking	Goat, human, cow, lion	Legs
Jumping	Kangaroo, monkey, frog	Legs
Flying	Bird, housefly, bee	Wings
Swimming	Fish, whale, dolphin	Fins or flippers and tail
Crawling	Crab, lizard, caterpillar	Legs or body muscles
Climbing	Monkey, squirrel, koala	Hands and legs

Q42. Why do animals have different ways of moving?

Answer: Animals have different body structures and live in different habitats. Their ways of moving help them find food, escape from predators and survive in their environments.

Q43. Name the Indian botanist mentioned in the chapter who worked on plant diversity.

Answer: Janaki Ammal, an Indian botanist, contributed to work related to plant diversity and conservation.

Topic 9: Plants and Animals in Different Surroundings

Q44. What is a habitat?

Answer: A habitat is the natural place where an organism lives and gets food, water, shelter and suitable conditions for survival.

Q45. What are the main types of habitats?

Answer: The three main types described in this chapter are terrestrial habitats, aquatic habitats and arboreal habitats.

Q46. What is a terrestrial habitat? Give examples.

Answer: A terrestrial habitat is a habitat found on land, such as forests, grasslands, deserts and mountains.

Examples of terrestrial animals include lions, tigers, camels and elephants.

Q47. What is an aquatic habitat? Give examples.

Answer: An aquatic habitat is a habitat in water, such as ponds, lakes, rivers, seas and oceans.

Examples of aquatic animals include fish, dolphins and whales.

Q48. What is an arboreal habitat? Give examples.

Answer: An arboreal habitat is a habitat in trees.

Examples of arboreal animals include monkeys and squirrels.

Q49. Where do the following animals commonly live?

Animal	Habitat
Bear	Forest or other suitable land habitat
Crocodile	Water and nearby land
Giraffe	Grassland or savanna
Camel	Desert
Squirrel	Trees and other suitable habitats
Tiger	Forest or grassland
Fish	Aquatic habitat
Turtle	Water or land, depending on the species
Elephant	Forest or grassland
Monkey	Trees and forest habitats

Q50. Can an organism live in more than one type of habitat?

Answer: Yes. Some organisms can live in more than one type of habitat. For example, frogs live on land and in water, while crocodiles use both water and land.

Topic 10: Habitat and Adaptation

Q51. What is adaptation?

Answer: Adaptation is a feature or change in the structure or behaviour of an organism that helps it survive in its habitat.

Q52. Why do organisms need adaptations?

Answer: Organisms need adaptations to obtain food and water, protect themselves, tolerate environmental conditions and reproduce successfully.

Q53. What are the basic needs provided by a habitat?

Answer: A habitat provides food, water, shelter, protection and a suitable place to live and reproduce.

Topic 11: Adaptations in Deserts

Q54. What are the main features of a desert habitat?

Answer: Deserts receive very little rainfall. They often have hot, dry conditions and a shortage of water.

Q55. How is a camel adapted to living in a desert?

Answer: A camel has long legs, broad padded feet, long eyelashes, closable nostrils and a hump that stores fat. It also conserves water by producing a small amount of urine.

Q56. What is the function of a camel's hump?

Answer: The hump stores fat, which can be used as an energy reserve when food is scarce.

Q57. How do a camel's feet help it walk on sand?

Answer: Its broad, padded feet spread its weight and help prevent it from sinking deeply into soft sand.

Q58. How does a camel protect itself from sand?

Answer: Its long eyelashes help keep sand out of its eyes, and its nostrils can close to prevent sand from entering.

Topic 12: Adaptations in Cactus Plants

Q59. What are xerophytes?

Answer: Xerophytes are plants adapted to survive in dry environments with little water.

Q60. How is a cactus adapted to a desert habitat?

Answer: A cactus has the following adaptations:

1. Its leaves are reduced to spines, which reduce water loss.
2. Its green stem makes food through photosynthesis.
3. Its thick stem stores water.
4. A waxy outer layer reduces water loss.
5. Its long roots can reach water deep in the soil.

Q61. Why are cactus leaves changed into spines?

Answer: The leaves are changed into spines to reduce water loss through transpiration and to provide some protection from animals.

Q62. How does the green stem of a cactus help it?

Answer: The green stem carries out photosynthesis and helps the plant make food.

Q63. Why does a cactus have a thick, waxy stem?

Answer: The thick stem stores water, while the waxy layer reduces water loss.

Topic 13: Adaptations in Mountains

Q64. What are the conditions in mountain habitats?

Answer: Mountain habitats are often cold and windy. The ground may be covered with snow, and plants may find it difficult to obtain water and nutrients.

Q65. How is a mountain goat adapted to its habitat?

Answer: A mountain goat has strong, rough and flexible hooves that help it climb rocky slopes. Its thick fur keeps it warm, and its strong legs help it jump between rocks.

Q66. How does the fur of a mountain goat help it?

Answer: Its thick fur protects it from cold weather and helps keep its body warm.

Q67. How do mountain goats protect themselves from predators?

Answer: Their horns help protect them from predators, while their agility helps them escape across rocky slopes.

Q68. How are plants adapted to mountain habitats?

Answer:

- Many mountain plants have short stems to withstand strong winds.
- Trees often have cone-shaped branches that allow snow to slide off.
- Needle-like leaves reduce water loss.
- Some trees remain green throughout the year.
- Thick bark helps protect trees from cold conditions.

Q69. Why do some mountain trees have cone-shaped branches?

Answer: Their cone-shaped branches allow snow to slide off easily and reduce damage to the branches.

Topic 14: Adaptations in Freshwater Habitats

Q70. What is a freshwater habitat?

Answer: A freshwater habitat is a habitat found in water with a low amount of dissolved salts, such as ponds, lakes and rivers.

Q71. How is a fish adapted to living in water?

Answer: A fish has a streamlined body, fins and a tail for swimming, and gills for breathing underwater.

Q72. What is a streamlined body?

Answer: A streamlined body is a shape that allows an organism to move through water or air with less resistance.

Q73. What are amphibians? Give an example.

Answer: Amphibians are animals that can live both on land and in water. A frog is an example.

Q74. How is a frog adapted to its habitat?

Answer: A frog has webbed feet for swimming, strong hind legs for jumping and moist skin that helps it exchange gases. Its lungs help it breathe on land.

Q75. How are aquatic plants adapted to life in water?

Answer:

- Their roots are often reduced because water is readily available.
- Their stems may be long, hollow and flexible.
- Floating plants may have broad leaves.
- Submerged plants often have narrow leaves that bend with water movement.

Q76. How do floating leaves of aquatic plants remain protected from water?

Answer: Many floating leaves have a waxy coating that helps repel water and protects them from decay.

Q77. What are the adaptations of Hydrilla?

Answer: Hydrilla is a submerged aquatic plant. It has thin leaves and flexible stems that help it remain underwater and move with water currents.

Topic 15: Saltwater Habitats

Q78. What is a saltwater habitat?

Answer: A saltwater habitat is a habitat containing water with a high concentration of dissolved salts, such as seas and oceans.

Q79. Name some animals found in saltwater habitats.

Answer: Fish, dolphins, whales, molluscs and sea anemones are found in saltwater habitats.

Q80. How do dolphins and whales breathe?

Answer: Dolphins and whales breathe air through blowholes on the upper parts of their heads. They must come to the water's surface to breathe.

Q81. How are dolphins and whales adapted to aquatic life?

Answer: They have streamlined bodies for swimming, flippers for movement and blowholes for breathing air. They can remain underwater for some time before coming to the surface to breathe.

Q82. How does a sea anemone protect itself?

Answer: A sea anemone attaches itself to rocks using its sucker-like base. This helps it remain attached despite water currents.

Q83. What are algae?

Answer: Algae are organisms that commonly live in water or moist environments. Many algae contain chlorophyll and make food through photosynthesis.

Q84. How do some algae attach themselves to rocks?

Answer: Some algae have root-like structures called holdfasts that help them attach to rocks and other surfaces.

3. Important activities from the chapter

Activity 1: Types of plants — Page 3

The textbook asks students to observe plants around their homes or school and classify them by height and use.

Name of plant	Category	Approximate height	Use
Coriander	Herb	Short	Used as food and seasoning
Rose	Shrub	Medium	Ornamental plant
Neem	Tree	Tall	Shade and other useful products
Watermelon	Creeper	Spreads along the ground	Edible fruit
Money plant	Climber	Varies with support	Ornamental plant

Note: Heights depend on the variety and growing conditions.

Activity 2: Leaf venation and roots — Page 5

Plant	Leaf venation	Root system
Mango	Reticulate	Taproot
Mustard	Reticulate	Taproot
Grass	Parallel	Fibrous
Wheat	Parallel	Fibrous
Pea	Reticulate	Taproot
Banana	Parallel	Fibrous

These are the usual patterns taught in the chapter.

Activity 3: Identify the habitat — Page 7

The chapter asks students to group animals according to where they live. The following is a sample answer.

Desert	Trees	Sea	Forest
Camel	Squirrel	Fish	Tiger
—	Monkey	—	Bear
—	—	—	Elephant

Some animals, such as crocodiles and turtles, can use more than one habitat. Their exact habitat depends on the species.

4. Multiple-choice questions (MCQs)

Chapter practice quiz

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Choose the correct option and check your answers at the end.

1. What is biodiversity?

A. Only the number of trees
B. The variety of living organisms
C. Only animals in a forest
D. Only plants in a garden

2. Which of the following is a herb?

A. Neem
B. Rose
C. Coriander
D. Mango

3. Which plant is a shrub?

A. Rose
B. Wheat
C. Grass
D. Watermelon

4. Which plant is a creeper?

A. Neem
B. Watermelon
C. Money plant
D. Rose

5. Which plant is a climber?

A. Money plant
B. Coriander
C. Neem
D. Wheat

6. What is reticulate venation?

A. Veins form a network
B. Veins are absent
C. Veins run parallel
D. Leaves have no veins

7. Which plant generally has fibrous roots?

A. Carrot
B. Mustard
C. Wheat
D. Pea

8. How many cotyledons does a monocot seed have?

A. One
B. Two
C. Three
D. None

9. Which is an example of a dicot plant?

A. Rice
B. Wheat
C. Maize
D. Pea

10. Which body part helps fish breathe underwater?

A. Lungs only
B. Gills
C. Fur
D. Horns

11. What is a habitat?

A. A type of food
B. The natural place where an organism lives
C. A type of root
D. A plant disease

12. Which animal is adapted to desert life?

A. Camel
B. Polar bear
C. Frog
D. Trout

13. What does a camel's hump store?

A. Water
B. Air
C. Fat
D. Sand

14. Why does a cactus have spines?

A. To increase water loss
B. To reduce water loss
C. To swim
D. To absorb saltwater

15. Which animal is an amphibian?

A. Camel
B. Frog

C. Giraffe
D. Housefly

16. What helps a mountain goat grip rocky slopes?

A. Webbed wings
B. Rough, flexible hooves
C. Gills
D. A streamlined body

17. What is an arboreal habitat?

A. A habitat in trees
B. A habitat in the sea
C. A habitat under ice
D. A desert habitat only

18. How do whales breathe?

A. Through gills
B. Through their skin only
C. Through blowholes
D. Through fins

19. What is a holdfast?

A. A structure that helps some algae attach to surfaces
B. A type of seed
C. A camel's hump
D. A fish's fin

20. What is a xerophyte?

A. A plant adapted to dry conditions
B. An animal living in water
C. A plant with no roots
D. A type of fish
Check answers

5. Fill in the blanks

1. The variety of living organisms is called biodiversity.
2. Short plants with soft stems are called herbs.
3. Medium-sized woody plants are called shrubs.
4. Plants with weak stems that spread along the ground are called creepers.
5. The arrangement of veins in a leaf is called venation.

6. A seed with one cotyledon belongs to a monocot plant.
7. A habitat is the natural place where an organism lives.
8. A camel's hump stores fat.
9. Cactus leaves are modified into spines.
10. Fish breathe underwater with the help of gills.
11. Animals that live both on land and in water are called amphibians.
12. Plants adapted to dry conditions are called xerophytes.
13. Dolphins and whales breathe through blowholes.
14. The root-like attachment structure of some algae is called a holdfast.
15. A network of veins in a leaf is called reticulate venation.

6. True or False

Statement	Answer
1. All plants have the same height and structure.	False
2. Rose is a shrub.	True
3. Watermelon is a creeper.	True
4. Wheat generally has fibrous roots.	True
5. Monocot seeds have two cotyledons.	False
6. Fish use gills to breathe underwater.	True
7. A camel's hump stores fat.	True
8. Cactus spines help reduce water loss.	True
9. A frog is an amphibian.	True
10. All aquatic animals breathe through gills.	False
11. Whales breathe air through blowholes.	True
12. A habitat provides food, water and shelter.	True

7. Quick revision notes

- Biodiversity: Variety of living organisms.
- Herbs: Short plants with soft stems.
- Shrubs: Bushy plants with woody stems.
- Trees: Tall plants with thick, woody trunks.
- Creepers: Plants that spread along the ground.
- Climbers: Plants that grow upwards with support.
- Reticulate venation: Veins form a network.
- Parallel venation: Veins run parallel to each other.
- Taproot: One main root with smaller branches.
- Fibrous roots: Many thin roots of similar size.
- Monocots: Seeds with one cotyledon.
- Dicots: Seeds with two cotyledons.
- Habitat: The natural home of an organism.

- Adaptation: A feature that helps an organism survive.
- Desert adaptation: Camels conserve water; cacti store water and have spines.
- Mountain adaptation: Thick fur and strong hooves help mountain goats survive.
- Aquatic adaptation: Streamlined bodies and fins help fish swim.
- Amphibians: Animals such as frogs that can live on land and in water.
- Saltwater adaptation: Dolphins and whales breathe air through blowholes.

These notes cover the main topics, visible textbook questions, identification activity and practical classification activities in the uploaded chapter. The extra questions and exercises are study questions prepared from the chapter content, rather than a claim that the textbook contains these exact questions.